

LONDON DATA WEEK

● HACKATHON · 2026

AI helped you train
your ML model —
but can you vouch for it?

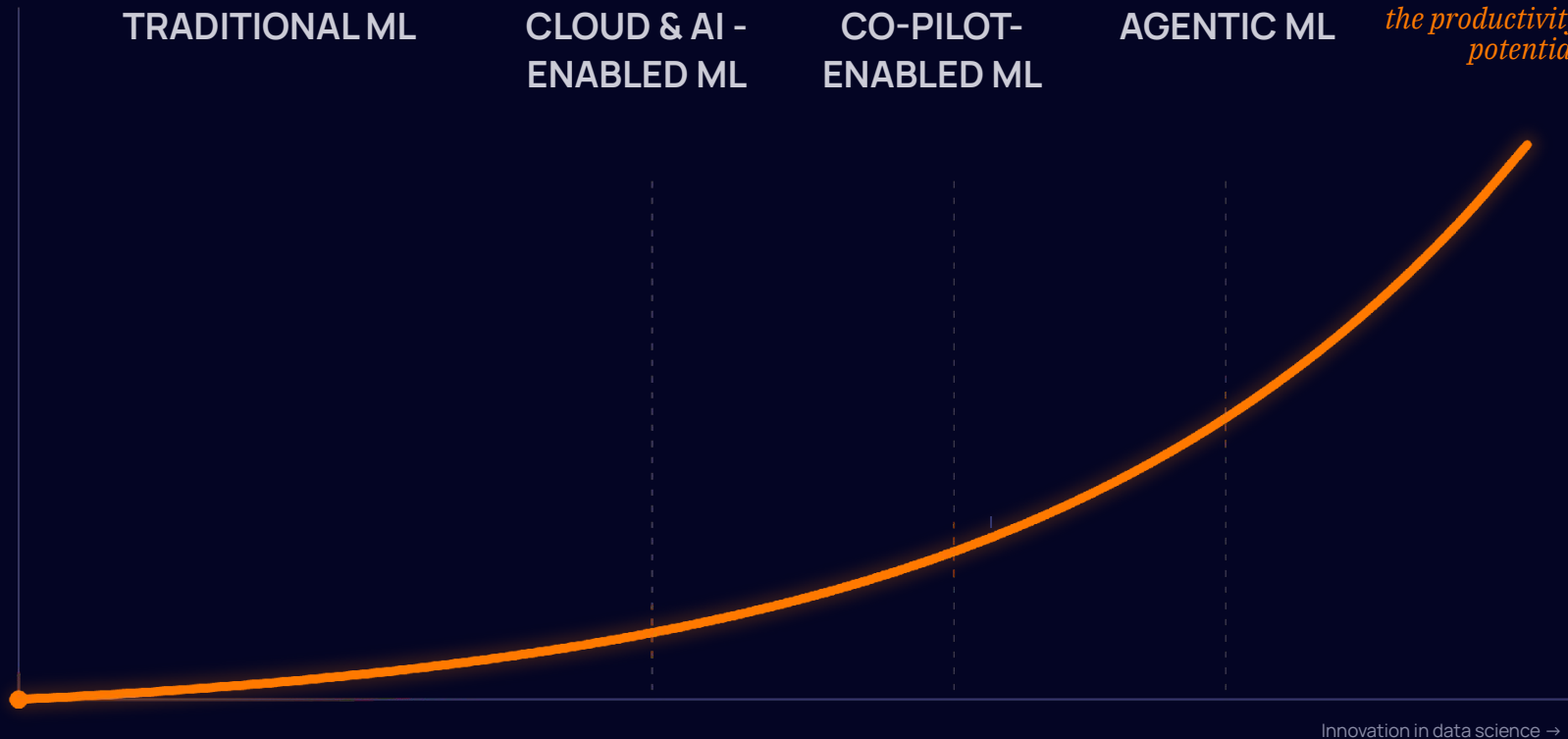
Co-hosted by Probabl and the GLA Data for London team.



 probabl.

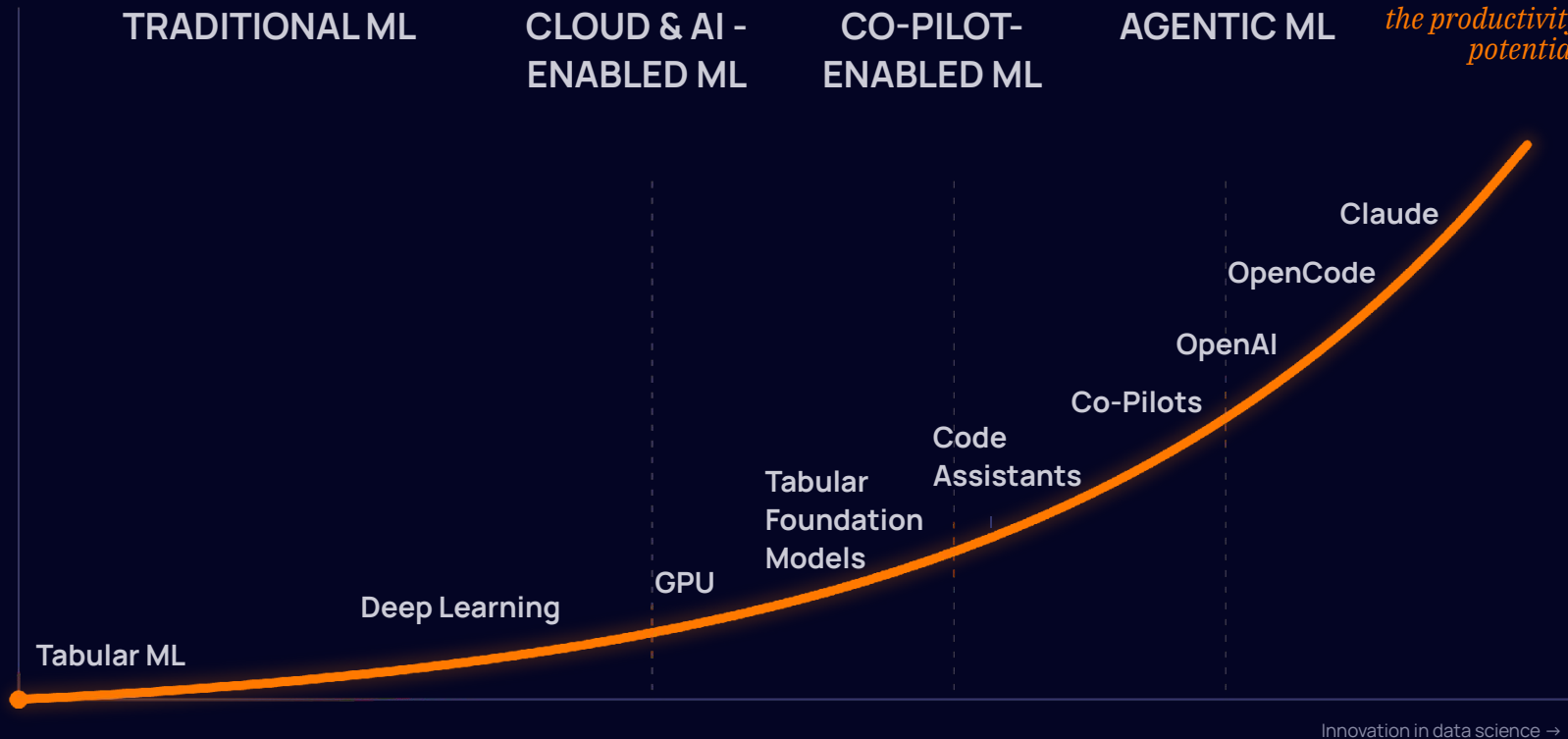
Innovation in data science *compounds*.

:proba1.
*the productivity
potential*

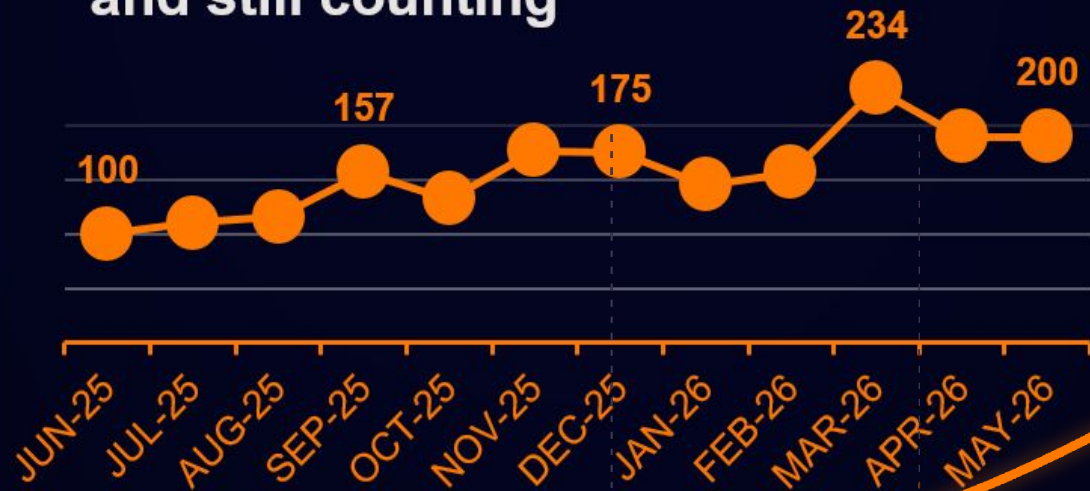


Innovation in data science *compounds*.

:probabl.
the productivity potential



Billions of downloads and still counting



:probabl.

02 WHY WE'RE HERE

Accelerate decisions,
without compromising
the analysis.



The plan for *today*.

9:30 - 10:00

1

Welcome &
team formation



10:00 - 12:00

2

Hackathon



12:00 - 13:00

3

Presentations
& prizes

Three *real London* problems.

1

Youth opportunity

2

Decarbonisation
opportunities

3

London data discovery

Pick one. Teams tackle their chosen problem with the datasets provided — assessed on how well you explain and evaluate your decisions.

Young Londoners Not in Education, Employment, or Training

 **1 million+**

young people NEET in the UK

 **19.3%**

London youth unemployment
(Q4 2024, above UK average)

 **Disparities**

Outcomes vary by ethnicity, disadvantage,
and disability

Problem: How might we tailor the Youth Offer for London to help the most young people into work or training?



Starting dataset: [KS4 leavers destinations](#) on Explore Education Statistics

End of KS4 = GCSEs (age 16). Tracks destinations in the first year after GCSEs. 4 sub-datasets; significant filtering needed.

You're welcome to explore other datasets on the same site, ONS, NOMIS, or any open-access official statistics.



Using data to identify decarbonisation opportunities



1 LAEP DataHub

Access 180+ public datasets from London's local area energy plans



5 StoryMaps

Combine narrative with spatial data by topics including solar, retrofit, heat networks, EV etc.,



Variations

Data across London varies due to different methodologies and reporting years

Problem: How might we identify buildings with high retrofit potential, particularly residential buildings, and particularly social housing?



Starting dataset: [London Building Stock Model v2](#) on the LAEP Datahub

You're welcome to explore other datasets on the LAEP Datahub, or any open-access official statistics.

Techniques: Geographic Information Systems, data linking



Discovering London's data

 20,000+

Datasets on the Data for London Library

 1,200+

Active users in 2025-26

 21.2%

Of total users visit the site using a mobile

Problem: How might we improve the Data for London Library search by enriching the metadata in its catalogue?



Starting dataset: [Data for London Library catalogue](#) available through the CKAN API endpoints available from [dfl-library.london.gov.uk](#)

You're welcome to explore other datasets from open data portals, including the London Datastore.



Techniques: Unsupervised learning, natural language processing

Form your *team*.

1

2-4 people

2

Choose a team
name

3

Pick a policy
problem

Get *building*.

EVERYTHING YOU NEED



Stuck?

Ask a Probablr to help!

Submit before 12:00

Data science skills for coding agents

A skill pack that brings scikit-learn, skrub, and skore methodology into any coding agent — so it can evaluate, explain, and vouch for what it builds.



Three open-source libraries

`scikit-learn`

The backbone of data science in python, developed and maintained by Probabl

Models, metrics, cross-validation

`skrub`

Extends the scikit-learn pipeline with preprocessing and a DataOps graph: from raw tables to a learner.

TableVectorizer, tabular_pipeline

`skore`

Makes evaluation straightforward: one call produces a structured report with metrics, visualizations, and checks.

skore.evaluate → report

👤 INSTALLATION

Installing in your preferred agent

```
$ pip install skore-cli  
$ skore skills install
```

Via skore CLI

Install the CLI with `pip/conda/uv/pixi/...`
then `skore skills install` drops the pack
into your agent.

Via npx

`npx skills add`
`github.com/probabl-ai/skills`
does the same.

In Claude Code

The repo is also a native Claude Code plugin
marketplace.



Works in any agentic context:
Claude Code, OpenCode, Cursor, Bob...

A pack of 13 skills

01 · ML pipeline lifecycle

Declare, evaluate, test, audit — one experiment, end to end.

`build-ml-pipeline`

`evaluate-ml-pipeline`

`test-ml-pipeline`

`smoke-test-ml-pipeline`

`audit-ml-pipeline`

02 · Iteration loop

Source the next experiment.

`iterate-ml-experiment`

`iterate-from-skore`

`iterate-from-user`

03 · Workspace & tooling

`organize-ml-workspace`

`python-code-style`

`python-env-manager`

`data-science-python-stack`

04 · API references

Read real signatures, don't hallucinate them.

`python-api`

The workflow

01

Install the pack

```
$ skore skills install
```

02

Describe the task

"Build a churn model on this CSV using the f1 score to evaluate."

03

Get a skore report

Structured evaluation with metrics, methodological checks and visualizations.

04

Iterate, audit, ship

Inspect the model, decide the next step to explore, stop when satisfied

Evaluate and inspect the generated model with skore

```
import skore
report = skore.evaluate(model, df, y)
report
```

Report for HistGradientBoostingRegressor	
Results	Checks Estimator Data Help
	?
Metric	HistGradientBoostingRegressor
R ²	0.923563
RMSE	37.231161
MAE	29.393199
MAPE	1.015896
Fit time (s)	0.872921
Predict time (s)	0.003382

Methodological checks

skore runs methodological checks and flags issues before you trust the score. Here, skore suggests tuning some hyperparameters left at default.

```
report.checks.summarize()
```

Checks summary (fast mode): 1 issue(s), 1 tip(s), 2 passed, 7 not applicable, 4 skipped, 0 ignored.

Issues (1) Tips (1) Passed (2) Not Applicable (7) Skipped (4) Ignored (0) ?

- [\[SKD016\]](#) **Estimator not tuned.** Estimator(s) left at default settings; consider tuning: ['learning_rate', 'max_leaf_nodes'] for HistGradientBoostingRegressor.

NOTE

Fast mode is on: expensive checks are skipped unless already cached.

Mute a check by passing its code to ignore, e.g. `.checks.summarize(ignore=['SKD001'])`.

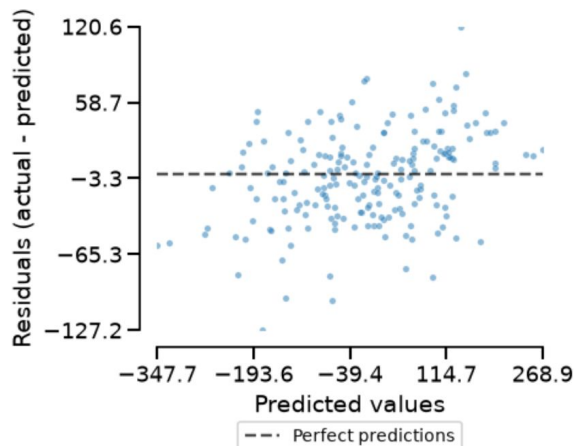
SKORE REPORT · DISPLAYS

Visualization Displays

skore lets you generate beautiful
visualizations quickly to get insight
that matter.

```
report.metrics.prediction_error()
```

Prediction Error for HistGradientBoostingRegressor
Data source: Test set

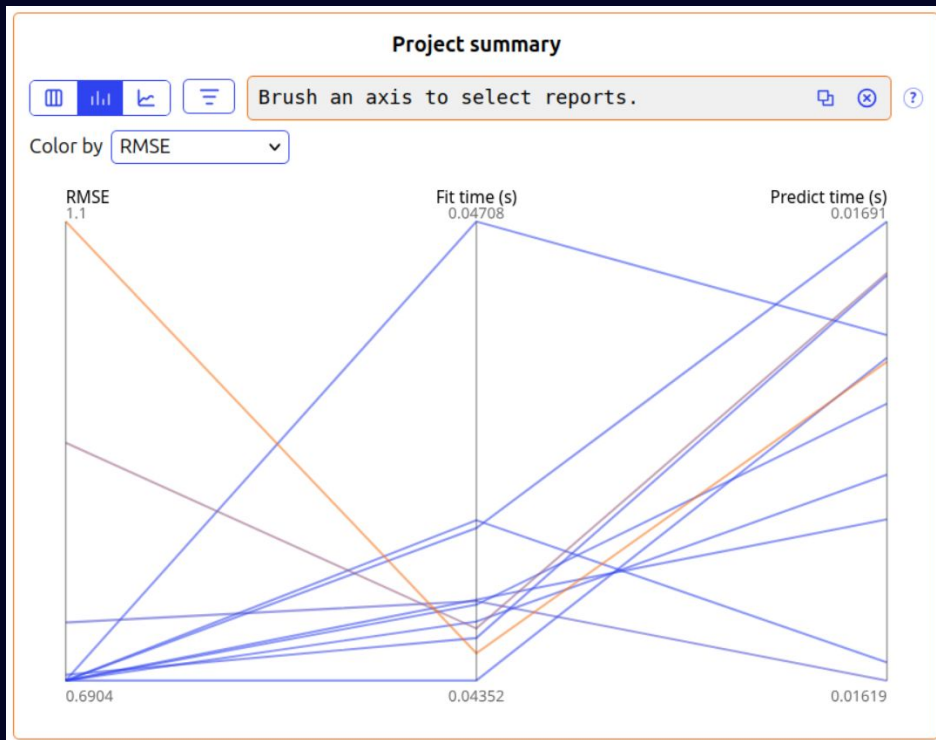


SKORE PROJECT · SUMMARY

Experiment tracking

Store your projects locally or on the hub, retrieve them at any time, pick the best one

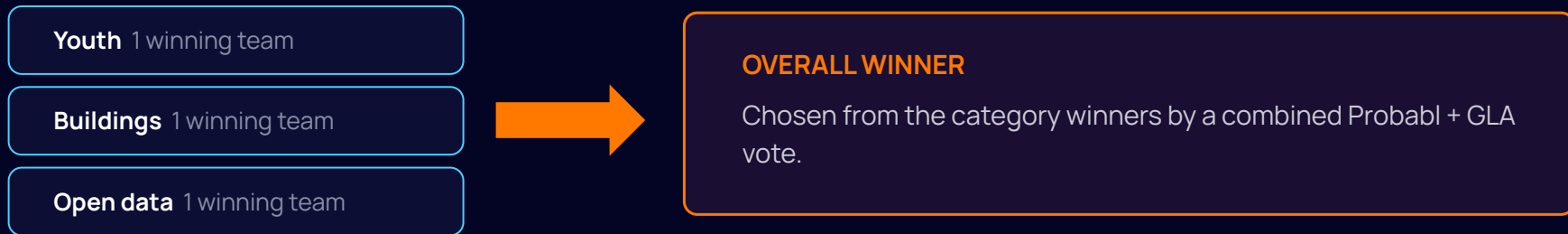
```
project.summarize()
```





How you'll be *judged*.

One winner per policy problem — a GLA + Probabl rep decides each.



WHAT MATTERS MOST

01 Methodology

02 Engagement with the issue

Higher leaderboard scores are encouraged — but they won't decide your place.



Prizes & *perks*.

OVERALL WINNER

- **Pitch to leadership** — Present to the GLA's data leadership team this afternoon
- **Probabl tote bags**

OVERALL WINNER & ALL CATEGORY WINNERS

- **100% discount** — Skolar certification

EVERY PARTICIPANT

100% discount code for the Skolar
Discovery badge.



Good luck —

and thank you for building with us.

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🔴 ONE LAST THING

How's your scikit-learn general culture?

Take the Discovery Badge — a short quiz on scikit-learn and ML fundamentals. Scan to start.

Coupon [LondonDataWeek2026](#)

100% off · valid all July · one use per participant



SCAN TO PLAY

*Please take a moment to complete the
feedback survey.*

!We value your feedback.

